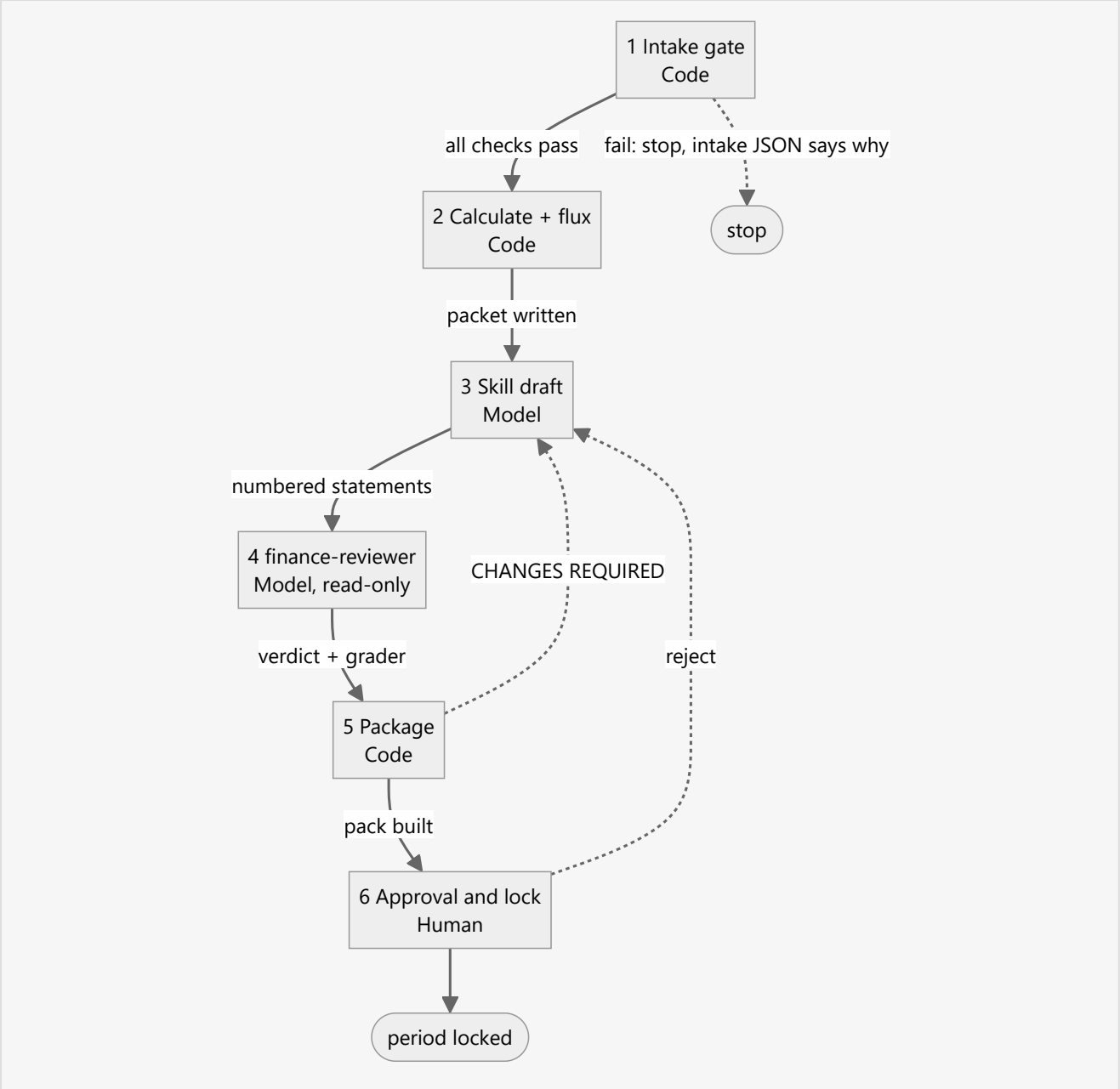
Skill and Analysis Workflow

How a client's monthly workbook becomes a reviewed analysis, what the skill is made of, how a new client is added, and what the output looks like.

End-to-end analysis flow

Six stages for one client and one month. Each stage has a gate: the next stage does not start until the gate passes. **Code** is deterministic Python, **Model** is Claude, **Human** is the reviewer.



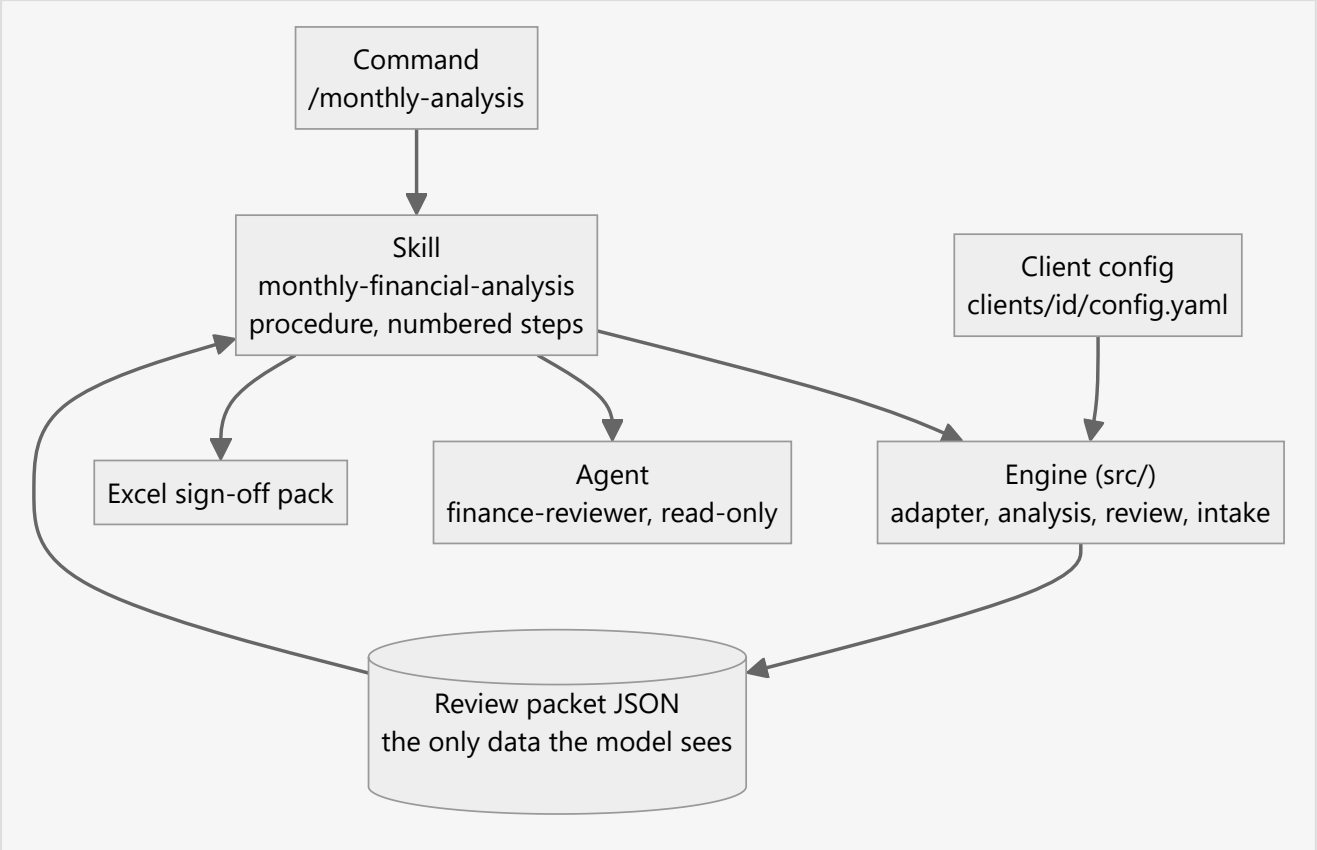
#	Stage	Gate
1	Intake gate	Workbook hash recorded; sheets, header, contiguous month columns and all total rows present; gross profit and operating income tie out; every section foots. Any failed check stops the run before any calculation
2	Calculate and flux	Runs in a client sandbox; months without enough history are "not available", never zero

#	Stage	Gate
3	Skill draft	Only numbers in the review packet; one numbered statement per line; no invented causes
4	finance-reviewer	Read-only agent returns `approve` or a numbered changes list
5	Package	APPROVED only if the reviewer said `approve` **and** the grader passed
6	Human approval	Reviewer name and reason required; approve refused unless the reviewer approved; an approved period is locked

Why numbers come from code and words from the model

A model asked to do arithmetic can be fluent and wrong. Here the model never sees the workbook, only the packet JSON for one client and month, and may state only numbers in it. Three independent checks then run on the text: the deterministic grader (every number, the 3 largest movements, cash and runway), the `finance-reviewer` agent (causes, direction words, unsupported claims), and a human.

Skill anatomy

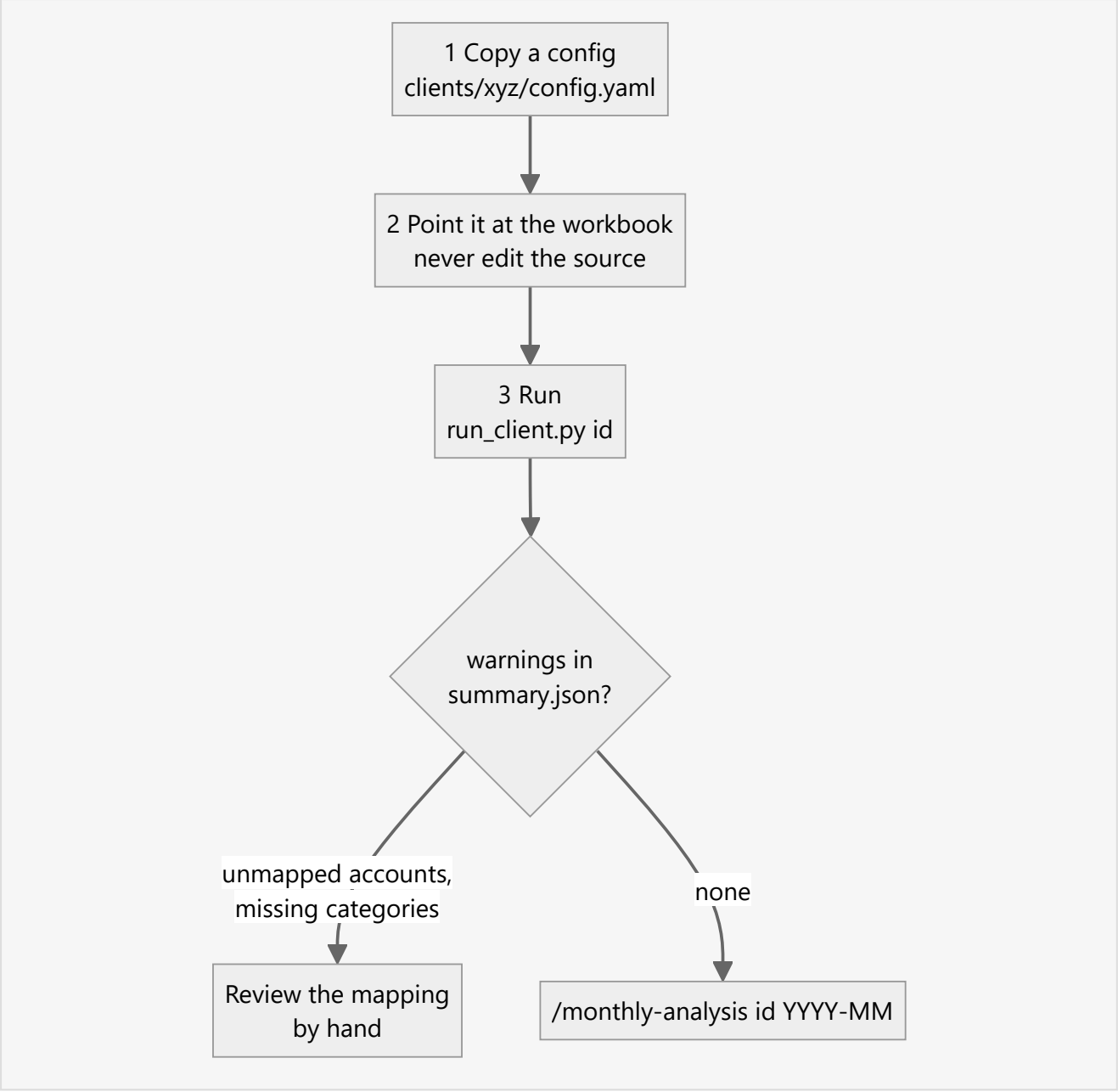


Piece	Role
Skill (`SKILL.md`)	The procedure: audit, revenue trend, margins, operating expenses, MoM rules, TTM, cash and runway, anomaly flags, commentary
Engine (`src/`)	Parses the workbook, runs the intake gate, computes every figure, writes the packet and the Excel pack
Config	Per-client facts: workbook path, sheet and label names, month format, account-to-category map, fiscal-year start, thresholds

Piece	Role
Agent	Read-only reviewer (Read, Grep, Glob only) that sees the packet and the commentary and nothing else
Command	`/monthly-analysis`: prepare, draft, review, revise once, build pack, report

Client-config model

A new client needs a config file, not code. The synthetic client `xyz` proves it: it differs from ABC in sheet name (`P&L`), header label (`Line Item`), month format (`%b-%y`, oldest first), section names, total names (`Net Profit`) and fiscal-year start (April). No engine code changed.



Limit: the parser expects a wide sheet with one column per month. A different layout needs code.

Sample output walkthrough (ABC, June 2026)

Input: the ABC workbook. The pipeline runs the intake gate, then writes the review packet, the flux table and the analysis workbook. The commentary is numbered, one statement per line:

#	Statement (abridged)
1	Revenue for 2026-06 was $195,898.16, a decrease of $12,452.80 (-5.98%) from 2026-05
2	Gross profit was $137,425.72 at a gross margin of 70.15%, up 3.11 percentage points
5–7	The three largest movements: Events & Conferences up $17,970.67; Bad Debts down $14,741.53 (from $12,935.29 to a credit of -$1,806.24); Usage Fees down $11,142.74
8	Trailing-twelve-month revenue was $2,303,975.51
9–10	Cash was $713,481.26; runway 3.40 months on a 3-month average burn of $209,656.62

The flux table flags 5 items for June 2026 (thresholds: $5,000 and 10%, sign flips over $1,000); Bad Debts is a `SIGN_FLIP`. The Excel sign-off pack has six sheets: Summary, Commentary Sign-off, Numbers Check, Top Movements, Reviewer Changes and Open Questions. The sign-off column is left **blank** for the human.

The engine's own text, the reviewed text and the saved skill output all pass the grader; the flawed control fails it.